

Design of a Wideband Millimeter-Wave CMOS Receiver Using a Reconfigurable Low-Noise Amplifier with 3-Winding Transformer Load

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Abstract—High-speed wireless communication systems require highly efficient and wideband receiver architectures capable of operating at millimeter-wave frequencies. This paper presents the design principles of a wideband CMOS receiver employing a reconfigurable low-noise amplifier (LNA) combined with a three-winding transformer load. The proposed architecture enables broadband operation, interference suppression, and on-chip image rejection. The receiver integrates RF amplification, frequency translation, and intermediate-frequency (IF) processing while maintaining high gain and bandwidth suitable for next-generation wireless systems. Fabrication results demonstrate the feasibility of implementing such architectures using advanced CMOS technologies.

Index Terms—Millimeter-wave receiver, low-noise amplifier, transformer load, broadband CMOS, RF mixers, wireless communication

I. INTRODUCTION

Fifth-generation (5G) wireless communication systems require extremely high data rates, low latency, and the ability to support a large number of connected devices. To meet these demands, communication systems increasingly utilize millimeter-wave (mm-wave) frequency bands where large bandwidths are available. [1]

Current 5G systems operate in both sub-6 GHz and millimeter-wave frequency ranges. The mm-wave bands include N257 (26.5–29.5 GHz), N258 (24.25–27.5 GHz), N259 (39.5–43.5 GHz), N260 (37–40 GHz), and N261 (27.5–28.35 GHz). These bands enable high-capacity wireless communication suitable for applications such as smart cities, autonomous vehicles, and immersive entertainment systems. [2]

However, designing receiver circuits at these frequencies introduces significant challenges including limited gain, high noise, and narrow bandwidth. To address these issues, this work investigates a wideband receiver architecture incorporating a reconfigurable LNA and a three-winding transformer load structure to enhance bandwidth and enable flexible operation.

The primary contributions of the architecture include wideband and reconfigurable operation, interference signal cancellation after the LNA stage, and on-chip image rejection implemented using a multi-stage filtering approach.

II. MODULATOR DESIGN AND IMPLEMENTATION

The proposed receiver architecture consists of several key components including a wideband low-noise amplifier (LNA), RF mixers, a local oscillator (LO) path chain, LO buffers, and intermediate-frequency (IF) circuits.

A key element of the design is the three-winding transformer load used within the LNA stage. Traditional transformer loads provide impedance transformation and bandwidth

enhancement; however, the three-winding transformer introduces additional coupling paths that improve gain distribution and allow multiple signal paths.

The gain of the system can be expressed using the impedance parameters of the transformer network:

$$Gain_1(j\omega) = g_m Z_{11}(j\omega) \quad (1)$$

$$Gain_2(j\omega) = g_m Z_{21}(j\omega) \quad (2)$$

$$Gain_3(j\omega) = g_m Z_{31}(j\omega) \quad (3)$$

where g_m is the transistor transconductance and Z_{ij} represents the impedance matrix of the multi-winding transformer network.

The LNA is designed to operate in a reconfigurable manner, enabling the receiver to adapt to different frequency bands. Following the LNA, the RF signal is downconverted using a g_m -boosted double-balanced mixer which improves conversion gain and suppresses unwanted signal components.

The LO path includes frequency generation and distribution circuits, followed by LO buffers that provide sufficient drive strength for the mixer stage. Finally, IF circuits perform additional signal processing and filtering before further baseband processing.

III. EXPERIMENTAL RESULTS AND SYSTEM DEMONSTRATION

The proposed receiver architecture was fabricated using a 22 nm CMOS fully depleted silicon-on-insulator (FDSOI) technology process. The fabricated chip integrates all major RF and IF blocks required for wideband receiver operation. [3]

Measurement results demonstrate the feasibility of the architecture for mm-wave applications. The three-winding transformer structure enables improved gain distribution and enhanced bandwidth performance compared with conventional transformer loads.

Additionally, the receiver supports interference suppression through post-LNA signal processing and incorporates on-chip image rejection through a multi-stage filtering structure. These features improve the overall robustness of the receiver in dense wireless environments.

The measured results confirm that the architecture provides wideband performance suitable for next-generation wireless communication systems.

IV. CONCLUSION

This paper presented the design of a wideband millimeter-wave CMOS receiver employing a reconfigurable low-noise amplifier with a three-winding transformer load. The architecture enables broadband operation, interference suppression, and on-chip image rejection. By integrating RF amplification, mixing, LO distribution, and IF processing within a single CMOS platform, the proposed receiver demonstrates the potential for efficient implementation in advanced wireless communication systems. Fabrication in 22 nm FDSOI CMOS confirms the feasibility of the design and highlights its applicability to emerging mm-wave technologies.

REFERENCES

- [1] M. Rajaei, "A Wideband Millimeter-Wave CMOS Receiver Using Reconfigurable Low-Noise Amplifier With a 3-Winding Transformer Load," Department of Electrical and Computer Engineering Seminar Presentation. [?]